ECON 6290: Topics in Micro Development (Labor Markets in Developing Countries)

Lisa Ho

Fall 2026
Week 1

Today's class

1. Introductions + About This Class
2. Topic 1: Human Capital

This class

- This class will focus (mostly) on labor economics topics in developing countries
  - Often with cross-cutting theme of gender
  - Also with sprinkling of miscellaneous topics that don't have another course home: health & development, political economy of development, climate & development,
- Fulfils requirements for development field for Economics PhD students at Columbia
  - Of general relevance to anyone interested in applied micro

Me (Lisa Ho)

- Grew up in Singapore and London before university in the US
- BS in Computer Science & Mathematical Economics (MIT) $\rightarrow$ MA in Global Affairs (Tsinghua) $\rightarrow$ PhD (MIT) $\rightarrow$ Postdoc (Yale) $\rightarrow$ Assistant Professor (Columbia)
- Research in development and labor economics
  - Job amenities (e.g. flexible work arrangements, parental leave)
  - Barriers to and impacts of female labor force participation
- Projects mostly in India (+1 in Egypt, +1 in Brazil, +1 in the United States)
  - Mostly field experiments with a couple of projects based only on admin data

You

1. Name, year, program
2. Where you’re from (interpreted as you wish)
3. What research topics you’re interested in
4. A work thing and a non-work fun thing you did this summer

A brief roadmap of the class

Development economics topics with a labor focus:

1. Human capital / education
2. Intrahousehold Issues - Resource Allocation, Decision Making, and Domestic Violence
3. Fertility and Marriage
4. Norms, Beliefs, and Attitudes
5. Reconciling Domestic and Market Work
6. Discrimination and Identity
7. “Behavioral” Labor Economics - Fairness, Identity, and Effort
8. Matching, Search, and Mobility Frictions
9. Labor Markets That Don’t Clear: Rigidity, Rationing, and Market Power
10. Health
11. Political Economy
12. Climate

Course goals and requirements (I)

- Class participation and reading responses (15%)
  - Mandatory paper reading for each week (*** on syllabus)
  - You'll submit a short response ahead of each class
  - In class we'll discuss the paper and our responses
  - Guidelines on reading responses are posted on website
- Each class one student will lead the discussion based on everyone's responses (10%)
  - Sign up form is posted on course website; please fill it out by next class
  - I'll lead discussion next week (on Duflo's South Africa cash transfers paper)
  - The week after, student-led discussions begin. Each student will lead 1-2 discussions.

Course goals and requirements (II)

- Referee reports (2 x 10%)
  - You will write two referee reports on recent job market papers related to development and labor (one RCT, one using observational data)
  - Papers to be selected from those on syllabus (use Drive link from syllabus!)
  - Non-RCT report is due Friday, Sep 18th; RCT report is due Friday, Sep 25th
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- Mini-Proposals (3 x 15%)
  - The primary goal of this class is to support you in the transition to writing a thesis
  - Each proposal asks you to come up with an idea within constraints
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  - These are short assignments; the grant proposals should be no longer than 2-3

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- Proposal Presentation (10%)
  - In the last class of the semester, you pick one of your mini-proposals to present

Any questions?

Today's class

1. Introductions + About This Class
2. **Topic 1: Human Capital**

Today's class

1. **What are the labor market returns to human capital?**
2. Education as an investment decision
3. Do people respond to the returns?
4. Does schooling actually produce human capital?
5. Are tests measuring what we think they are?

A canonical question in labor economics

- **What are the labor market returns to human capital?**

- One of the most estimated parameters in all of economics

- Why is it important?
  - Input into how much individuals should invest in human capital
  - Input into government decisions about about school building & other policies

A measurement problem

- **Human capital is really hard to measure**
  - Also ambiguously defined
- So in practice economists often use **years of schooling** as a stand in
  - Observable, comparable across people, collected in essentially every household survey

The standard returns-to-schooling regression

- Standard Mincer equation:

$$\ln w_i = \alpha + \beta S_i + \gamma X_i + \epsilon_i$$

  - w_i is the wage of person i, S_i is years of schooling
  - X_i is whatever else we control for (often experience and its square)

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Psacharopoulos & Patrinos (2018): 1,100 estimates, 139 countries

	Return per year (%)	Mean years of schooling
Latin America & Caribbean	11.0	7.3
Sub-Saharan Africa	10.5	5.2
East Asia & Pacific	8.7	6.9
South Asia	8.1	4.9
Europe & Central Asia	7.3	9.1
Middle East & North Africa	5.7	7.5
World average	**8.8**	**8.0**

- Returns are higher in low-income countries
- Returns are $\approx$ 2 pp higher for women than men

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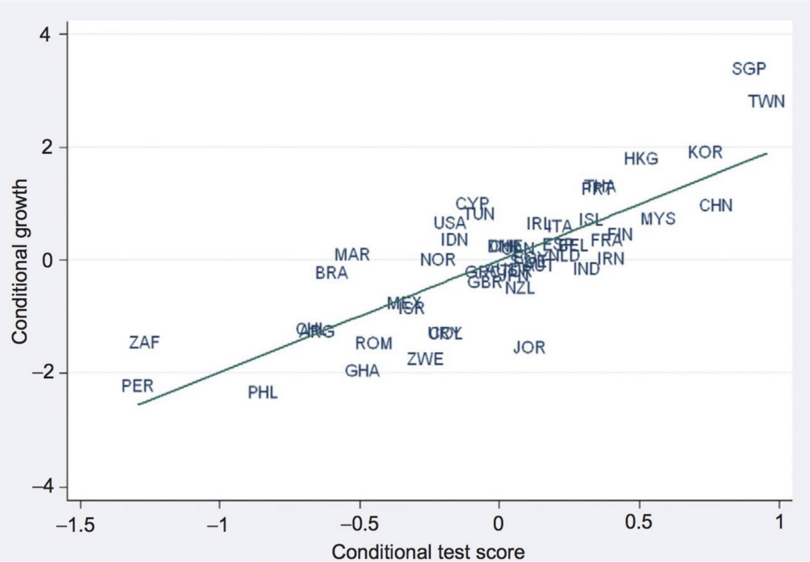
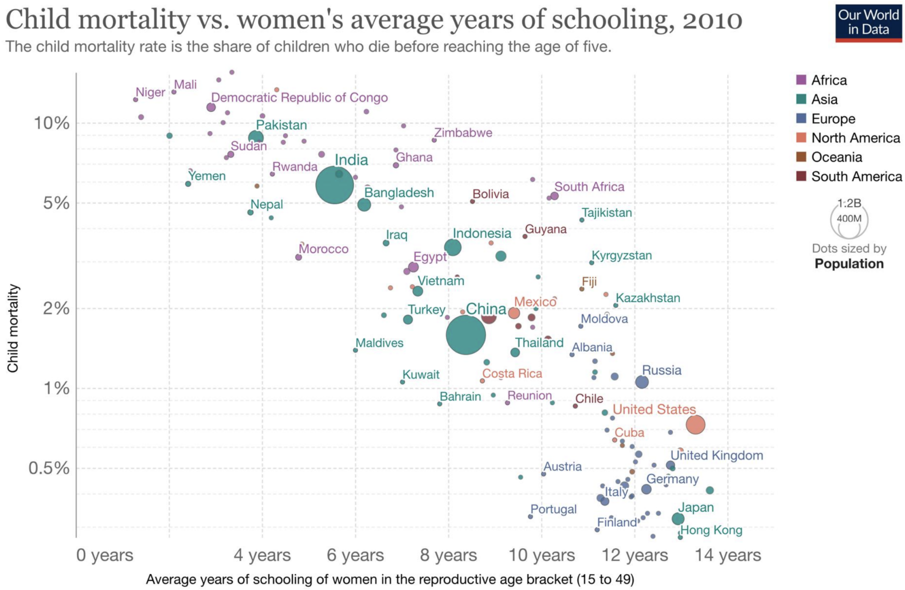
But... Mincerian OLS estimates are unlikely to be causal

At the country level, education is correlated with economic growth

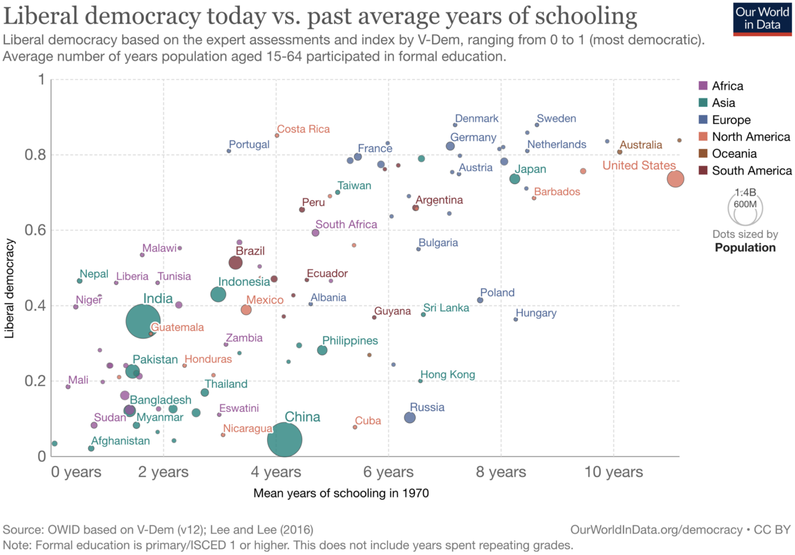


Hanushek & Woessmann 2008

And with child mortality



And with democratic values…



Many reasons OLS does not give us causal effect of schooling on earnings

- **“Ability” bias**: whatever makes you good at school may also make you good at earning. S_i is correlated with ϵ_i
- **Family background**: richer, better-connected parents buy both more schooling and better jobs
- Your suggestions
- Every one of these biases the OLS estimate (and mostly makes returns to schooling look larger)

The Fundamental Problem of Causal Inference

- Consider evaluating the effect of an extra year of schooling on wages

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- Notation:
  - Y_i^T = wages of person *i* **with** the extra year of schooling
  - Y_i^C = wages of person *i* **without** it
  - Treatment effect for person *i*: $Y_i^T - Y_i^C$

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- **Fundamental problem**: We never observe the same person *i* both with and without that year of schooling
- What can we do?
  - We cannot know the return for a *particular* person
  - But we may hope to learn the **average treatment effect**: $E[Y_i^T - Y_i^C]$

The Selection Bias Problem

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- Lots of modern empirical methods to try to solve this problem

The Gold Standard: Randomized Experiments

Why Randomization Solves Selection Bias

$$
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- Regression counterpart:

$$Y_i = \alpha + D \cdot \mathbf{1}(i \in T) + \epsilon_i$$

where $\mathbf{1}(i \in T)$ is dummy for being in treatment group

But we are not going to randomly assign children to 6 vs 12 years of education

- So the credible estimates of the return to schooling come from somewhere else:
  - **Policy variation** — a government did something abrupt, to some people and not others
  - Compulsory schooling laws, school construction, changes in the school-leaving age

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- We will work through a canonical example in the development literature: Duflo in Indonesia

Duflo (2001): school construction in Indonesia

- **Setup**:
  - A very large, very fast primary school construction campaign (INPRES), financed by the 1973 oil boom
  - Over 61,000 primary schools built between 1973 and 1978
  - Intensity of construction depended on *pre-campaign* enrollment: more schools where enrollment was low

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- **Difference-in-differences**:
  - **Treated cohorts**: aged 2–6 in 1974 — young enough to benefit from a new school
  - **Control cohorts**: aged 12 or older in 1974 — too old to benefit
  - **Treated regions**: program intensity above median; **control regions**: below

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- Sources of variation then interacted: **when you were born** $\times$ **where you were born**

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- Psych! Parallel trends assumption is about a counterfactual; can't be tested directly
- But there are some checks we can do to increase our confidence
- **Placebo experiment**: run the same design on "old versus very old" cohorts — both too old to have been treated
  - If we find an "effect" there, high-intensity and low-intensity regions were already diverging
  - If we find nothing, we are more comfortable

Continuous treatment intensity across groups

- Suppose we have G groups and the intensity of treatment differs across them. Think of it as several treatments: $Y_i(w)$ for $w = 0, 1, 2, \ldots, G$
    - Or let the treatment take continuous values, i.e. number of schools built per 1000 kids
    - Control for district-of-birth dummies, and interact *post* with schools per 1,000 children

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- With only two cohorts:

$$Y_i = \alpha + \beta T_t + \sum_{g=1}^{G} \gamma \mathbf{1}[G_i = g] + \tau_C (S_g \cdot T_t) + \epsilon_i$$

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- With multiple periods but still one "pre" and one "post", replace T_t by year-of-birth dummies:

$$Y_i = \alpha + \sum_{t=1}^{T} \beta_t \mathbf{1}[T_i = t] + \sum_{g=1}^{G} \gamma_g \mathbf{1}[G_i = g] + \tau (S_g \cdot POST) + \epsilon_i$$

Duflo (2001): results

Duflo (2001)

Table 4—Effect of the Program on Education and Wages: Coefficients of the Interactions Between Cohort Dummies and the Number of Schools Constructed per 1,000 Children in the Region of Birth

		Dependent variable					
		Years of education			Log(hourly wage)		
	Observations	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Experiment of Interest: Individuals Aged 2 to 6 or 12 to 17 in 1974 (Youngest cohort: Individuals ages 2 to 6 in 1974)							
Whole sample	78,470	0.124	0.15	0.188			
		(0.0250)	(0.0260)	(0.0289)			
Sample of wage earners	31,061	0.196	0.199	0.259	0.0147	0.0172	0.0270
		(0.0424)	(0.0429)	(0.0499)	(0.00729)	(0.00737)	(0.00850)
Panel B: Control Experiment: Individuals Aged 12 to 24 in 1974 (Youngest cohort: Individuals ages 12 to 17 in 1974)							
Whole sample	78,488	0.0093	0.0176	0.0075			
		(0.0260)	(0.0271)	(0.0297)			
Sample of wage earners	30,225	0.012	0.024	0.079	0.0031	0.00399	0.0144
		(0.0474)	(0.0481)	(0.0555)	(0.00798)	(0.00809)	(0.00915)
Control variables:							
Year of birth*enrollment rate in 1971		No	Yes	Yes	No	Yes	Yes
Year of birth*water and sanitation program		No	No	Yes	No	No	Yes

Notes: All specifications include region of birth dummies, year of birth dummies, and interactions between the year of birth dummies and the number of children in the region of birth (in 1971). The number of observations listed applies to the specification in columns (1) and (4). Standard errors are in parentheses.

Intensity as an instrument to get returns to education

- So far we have the effect of *school construction* on schooling and earnings
- To get *return to education*, instrument schooling with program intensity $\times$ cohort

Intensity as an instrument to get returns to education

- So far we have the effect of *school construction* on schooling and earnings
- To get *return to education*, instrument schooling with program intensity $\times$ cohort
- **What can we use as instruments?**
  - **One**: program intensity in your district of birth, interacted with a dummy for being young enough to be exposed
  - **Many**: the full set of intensity $\times$ year-of-birth interactions — one per cohort. More power, and the first stage can vary with exposure

Duflo (2001): OLS and IV returns

Duflo (2001)

TABLE 7—EFFECT OF EDUCATION ON LABOR MARKET OUTCOMES: OLS AND 2SLS ESTIMATES

Method	Instrument	(1)	(2)	(3)	(4)
Panel A: Sample of Wage Earners					
Panel A1: Dependent variable: log(hourly wage)					
OLS		0.0776	0.0777	0.0767	
		(0.000620)	(0.000621)	(0.000646)	
2SLS	Year of birth dummies*program intensity in region of birth	0.0675	0.0809	0.106	0.0908
		(0.0280)	(0.0272)	(0.0222)	(0.0541)
		[0.96]	[0.9]	[0.93]	[0.9]
2SLS	(Aged 2–6 in 1974)*program intensity in region of birth	0.0752	0.0862	0.104	
		(0.0338)	(0.0336)	(0.0304)	
		(0.0338)	(0.0336)	(0.0304)	
Panel A2: Dependent variable: log(monthly earnings)					
OLS		0.0698	0.0698	0.0689	
		(0.000601)	(0.000602)	(0.000628)	
2SLS	Year of birth dummies*program intensity in region of birth	0.0756	0.0925	0.0913	0.134
		(0.0280)	(0.0278)	(0.0219)	(0.0631)
		[0.73]	[0.63]	[0.58]	[0.7]

What's the exclusion restriction and do we believe it?

What's the exclusion restriction and do we believe it?

- School-building choice must affect earnings *only* through years of schooling
  - But schools are also employers, and infrastructure, and a signal of government attention
- Also GE effects…
  - If building schools increases local supply of educated labor, may reduce skill premium
  - Then school building affects wages through others' schooling not just individua's own
  - Addressed in a follow up paper?

Whose return-to-education is this?

- The compliers for this instrument are the people whose schooling changed in response to the policy

- Who are they here?
  - Children in the **treated cohorts**, in **high-intensity districts**, who got more schooling *because a school was built near them*
  - We don't estimate the effect for children who would have gone anyway ("always takers")
  - We also don't estimate the effect for children who still did not go ("never takers")

- So the compliers are disproportionately children for whom physical access was the binding constraint

- May be lower or higher than the returns-to-education that a specific family is considering when deciding on educational investments

Education as an investment decision

1. A simple model

2. Empirics

A simple model

- A family chooses years of schooling S for a child

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- If the family can borrow and lend freely at interest rate r, the optimum is where

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- Keep investing while the return exceeds the interest rate & stop when they equalize

What this model predicts

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- With functioning credit markets, the family borrows against the child's future earnings, so:
    - **No income effect**: rich and poor families choose the same schooling
    - **Education levels converge**: there is no mechanism for schooling gaps to persist across generations

- And therefore: **there is no reason for education policy**
    - Families are already choosing correctly, given prices and returns
    - A subsidy just distorts a decision that was right to begin with

- Many economists believed a version of this for a long time: **the best education policy is no education policy**

But of course

- Most people (even economists) would say this is obviously not true
- To name a few reasons...
    - **Credit markets do not work like that.** A landless household cannot borrow against child's future wages
    - **Parents decide for children.** There is no reason the parent's optimum is the child's
    - **Nobody knows β.** Families act on *perceived* returns which may be wrong
    - **Schooling may not be only an investment**, e.g. can be consumption or status good

Still a useful lens

- The neoclassical model is wrong, but investment framing still useful
- It may still be right that families are weighing costs against returns, just
  - non-optimally, and
  - under constraints that are sometimes invisible/difficult to measure
- A common question that's been asked in development + labor:
- **Do people respond to the incentives of labor market returns when making educational investments?**

Do people respond to the returns?

1. Jensen (2010): do people even know the returns?
2. Dizon-Ross (2019): investment under real constraints

The (perceived) returns to education (Jensen, 2010)

- Dominican Republic. Ask eighth-grade boys what they think a secondary education pays
- **Measured** returns to schooling are high. **Perceived** returns are extremely low
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- **Measured** returns to schooling are high. **Perceived** returns are extremely low
- So even if following "invest until the return to another year equals the interest rate" the wrong number was used
- **Design**: randomly selected schools were given information on the measured returns

What they think school pays vs what it actually pays

	(1) Measured Mean	(2) Perceived (self)	(3) Perceived (others)
Not Complete Secondary	3,180	3,516	3,478
	[1400]	[846]	[829]
Complete Secondary	4,479	3,845	3,765
	[1432]	[1003]	[957]
(Sec) – (Not Sec)	1,299	329	287
		[406]	[373]

Jensen (2010)

Effects of the intervention on schooling

TABLE 3. EFFECTS OF THE INTERVENTION ON SCHOOLING

	Δ Implied Return (Self)		*Returned Next Year*			*Completed Secondary*		*Years of Schooling*	
	(1)	(2)	(3)	(4)	(5)	(7)	(8)	(9)	(10)
Treatment	366	366	.039	.041		.020	.023	.18	.20
	(29)	(29)	(.025)	(.023)		(.024)	(.020)	(.098)	(.083)
Log (income per capita)		30.0		.075			.21		.75
		(48)		(.042)			(.044)		(.16)
School Performance		1.1		.011			.019		.085
		(13)		(.010)			(.008)		(.035)
Father's education		-26		.082			.061		.28
		(33)		(.029)			(.029)		(.12)
Interviewed					.014				
					(.027)				
R^2	.087	.087	.002	.015	.0002	.001	.036	.003	.040
Observations	1,859	1,859	2,250	2,250	1,575	2,250	2,250	2,074	2,074

Notes: Heteroskedasticity-consistent standard errors accounting for clustering at the school-level in parentheses. *Δ Implied Return (Self)* is the change (round 2 minus round 1) in the implied returns to schooling the student expects for themselves (earnings with secondary completion minus earnings with primary only) in 2001 Dominican pesos (no adjustment made for inflation between the two rounds). School Performance is teacher assessment of the student's performance, on a scale of 1 to 5 (much worse than average, worse than average, average, above average, much better than average).

Jensen (2010)

Parents' beliefs about their own children (Dizon-Ross, 2019)

- Jensen asks whether people are miscalibrated on average about returns to schooling
- Dizon-Ross asks if parents are miscalibrated about their own children

Parents' beliefs about their own children (Dizon-Ross, 2019)

- Jensen asks whether people are miscalibrated on average about returns to schooling
- Dizon-Ross asks if parents are miscalibrated about their own children
- Design: Malawi, low-income parents with at least two school-age children
  - At baseline their beliefs about how each child is doing are often inaccurate
- Randomly selected parents are told verbally how each child actually performed
  - Delivered aloud, because low literacy levels
- Having two children was important so she could measure how parents allocate *between their own children*

Summary of the model in Dizon-Ross

- A parent chooses how much to invest in each child
- Suppose beliefs are **attenuated** i.e. they related to the truth but too flatly:

$$\text{believed}_i = a + b \cdot \text{true}_i, \qquad b < 1$$

  - Parents know roughly who is doing better, but understate how big the differences are
- Then investment, which responds to *beliefs*, also varies too flatly between kids
  - The slope of investment on true performance is attenuated by exactly that same b

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Same idea as illustrated in the paper

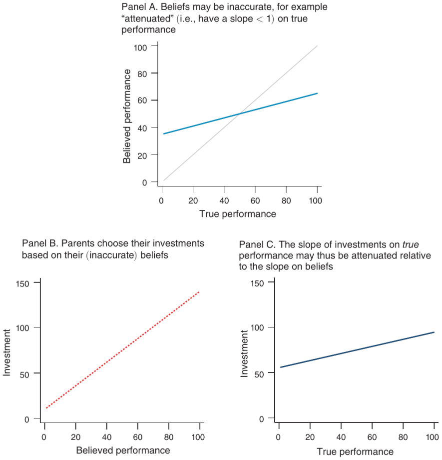

Figure 1. Empirical Approach: Inaccurate Beliefs about Performance Can Cause the Slope of Investments as a Function of Performance to Be Flatter than the Slope as a Function of Beliefs

Dizon-Ross (2019), Figure 1

Dizon-Ross: results

- Poor parents are more miscalibrated / less informed about their children's performance

- In line with prediction, parents update their beliefs and reallocate:
  - Increase enrollment of their higher-performing children
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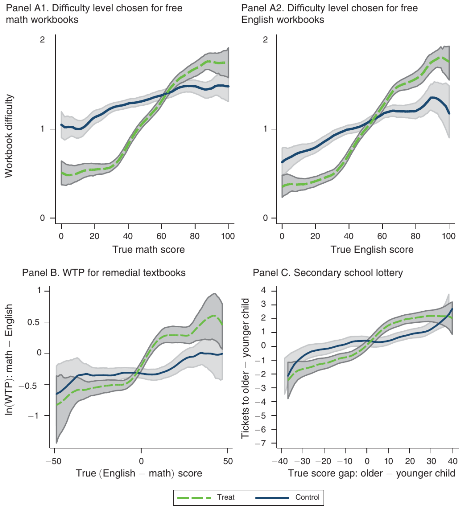
- In line with prediction, parents update their beliefs and reallocate:
  - Increase enrollment of their higher-performing children
  - Decrease enrollment of their lower-performing children
- Something uncomfortable about this?
  - May be efficient to maximize the household's return
  - But what about the children's welfare...?

Information steepens the investment slope



Dizon-Ross (2019), Figure 5. Dashed green = treatment, solid blue = control

Does schooling produce human capital?

1. The world got a lot more schooling
2. It did not get proportionally more learning

The substitution we made at the start

- Everything so far has used **years of schooling** as the measure of human capital

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The substitution we made at the start

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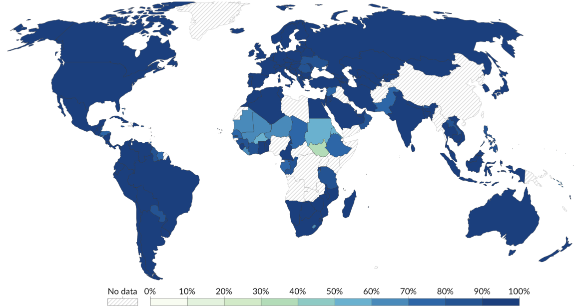
- Every return we estimated was a return to *schooling*, not to *skill*
- For the rest of class: **how good is that substitution?**
  - First, what has actually happened to schooling and to learning
  - Then, what makes schooling produce learning
  - Then, whether our measure of learning is any good either

Most children around the world are enrolled in school

% primary-school-age children in school, 2024



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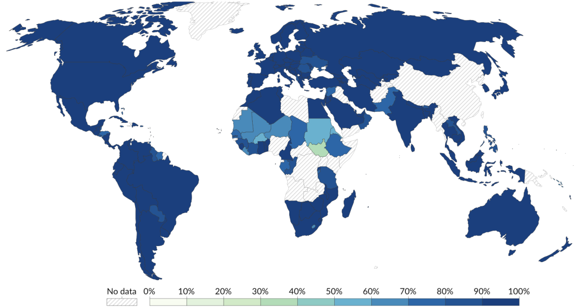
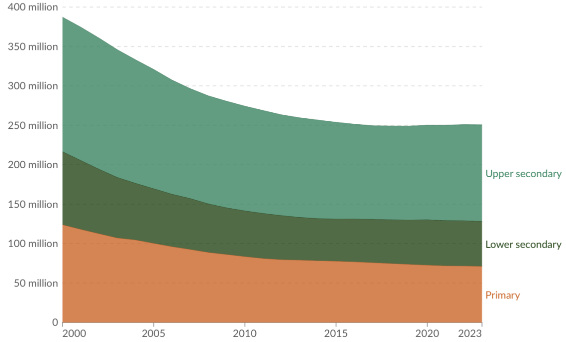


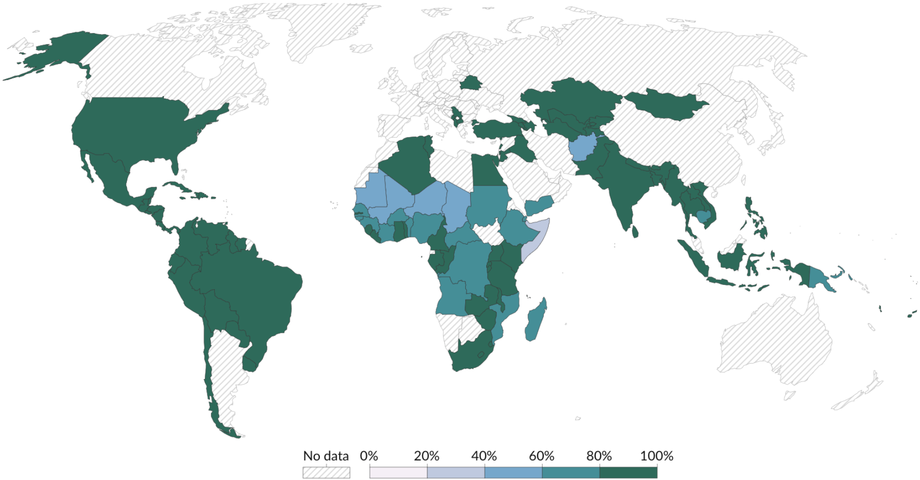
Number of children/adolescents not in school



Enrollment has risen enormously — and recently

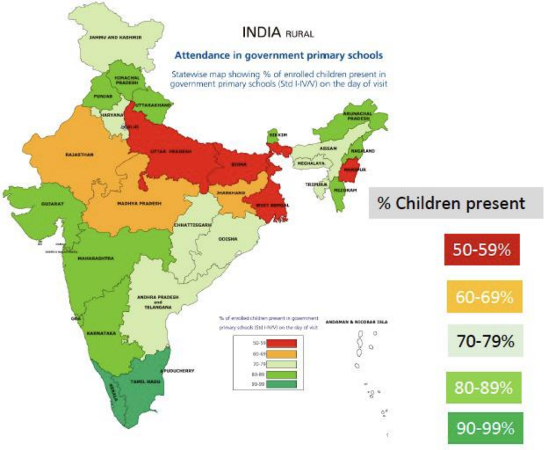
- Primary enrollment is close to universal in most of the world
- Secondary enrollment has risen sharply since the 1970s
- Tertiary enrollment is the fastest-growing of the three from a low base

Although they are absent a lot



Net attendance rate of primary school, 2024

Absenteeism, up close



And when they are in school, are they learning?

$$\begin{array}{r} 52 \\ -\ 24 \\ \hline \end{array} \qquad \begin{array}{r} 76 \\ -\ 47 \\ \hline \end{array}$$

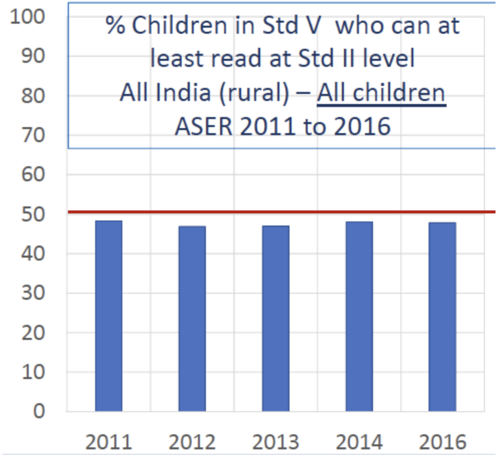
$$7\,\overline{)\,869\,}$$

All India (rural): All children	
ASER 2014 Grade	% Children who can do subtraction
Std III	25.3
Std IV	40.2
Std V	50.5

All India (rural): All children	
ASER 2014 Grade	% Children who can do division
Std V	26.1
Std VI	32.2
Std VIII	44.1

ASER, India

ASER over time



ASER, India

What works for increasing learning at school?

1. Remedying education (Banerjee, Cole, Duflo and Linden)
2. Peer effects, teacher incentives and tracking (Duflo, Dupas and Kremer)

"Teaching at the right level"

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- **Teaching at the Right Level (TaRL)**

- The diagnosis: in a typical classroom, the curriculum is pitched far above where most children actually are
  - The curriculum is set centrally and moves on schedule
  - Teachers teach to the top of the class, or to the syllabus
  - A child who fell behind in grade 2 sits through grades 3, 4 and 5 understanding almost nothing

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- The prescription: **group children by what they can currently do and teach to that**

- Two papers that between them make the case

Remedying education (Banerjee, Cole, Duflo and Linden, 2007)

- Two experiments with Pratham in urban slums, Mumbai and Vadodara

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- **Computer-assisted learning**: two hours a week of shared computer time playing games that drill mathematics
- Both are cheap, and neither adds a qualified teacher
    - The paper opens by noting that adding conventional school resources has mostly *not* worked

Remedying education: results

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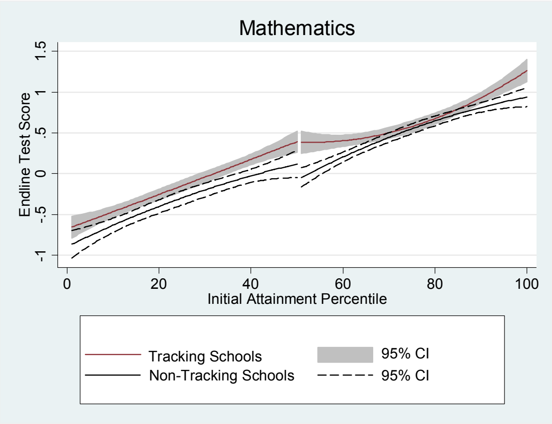
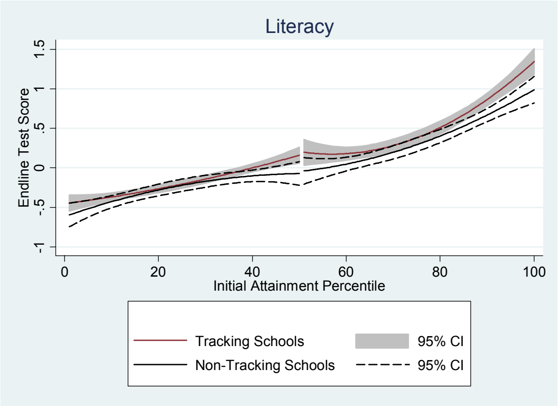
Endline scores across the initial attainment distribution





Duflo, Dupas and Kremer (2011), Figure 3

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  - **Tracking schools** split by initial achievement
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- Note class size is held fixed, so this is purely about peer composition
- **Unclear what directione ffects should go**
  - If you benefit from high-achieving peers, tracking helps the strong and hurts the weak
  - But if tracking lets teachers aim instruction at the actual level of the class, everyone can gain

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  - But for low achievers that loss was more than offset by teachers finally teaching at their level
- Lower-achieving pupils benefit most from tracking precisely when teachers otherwise have incentives to **teach to the top of the distribution**

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- Worth noticing: the **same intervention** has opposite sign predictions depending on how teachers respond. That is a general hazard in this literature

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- That is the whole idea, and it is now the basis of programs reaching millions of children

Are tests measuring learning?

1. Fictional money, real costs (Duquennois)
2. Cognitive endurance as human capital (Brown, Kaur, Kingdon and Schofield)

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- Two papers I like enormously, both of which make extremely creative use of data that already existed

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- Setting: standardised exams, including PISA. Some mathematics questions happen to be framed around **money**; others are not
- **Design**: compare how a student does on monetary versus highly similar non-monetary questions, and use the randomised ordering of questions to look at effects on *subsequent* questions

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- This paper asks whether **poverty itself** interferes with the measurement — and with the performance

- Setting: standardised exams, including PISA. Some mathematics questions happen to be framed around **money**; others are not

- **Design**: compare how a student does on monetary versus highly similar non-monetary questions, and use the randomised ordering of questions to look at effects on *subsequent* questions

- Note what the within-student comparison buys: no need to compare rich and poor children to each other at all

Matched monetary and non-monetary questions

Figure 6: Examples of Matched ASSISTments Questions

Kate went shopping with $72 in her pocket, but she didn't want to spend it all. She decided to spend 25% of her money at most, and save the rest for later. How much was Kate willing to spend?

David has 840 cookies. He decides to give 96% of them to a friend as a birthday present. How many cookies does David give away?

A charity is performing a fund raising campaign, below are the amounts of money raised each week:

$683, $1357, $352, $1946, $301, $1577

Calculate the **mean** dollar amount of money raised per week (round to nearest dollar).

Below are the number of spam emails filtered each week over the past few weeks by a school email system:

1073, 538, 964, 514, 273, 340

What is the **mean** number of spam emails filtered per week?

Duquennois (2022), Figure 6

Fictional money: results

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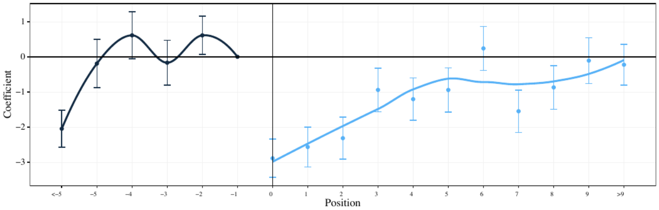
- Pair this with the cognitive endurance paper: in both, something we would normally call "ability" turns out to be partly a function of the conditions the child is sitting in

And it spills onto the questions that follow

Figure 9: Differential Performance by Position Relative to Monetary Event for Below National Median Students in TIMSS



Note: Data is limited to booklets that feature a single monetary question or booklets that feature only two sequential monetary questions. Estimating equation includes Student, Question, Below Med. x Diff., Below Med. x Seq., Below Med. x QType x Country, Below Med. x QTopic x Country fixed effects.

Duquennois (2022), Figure 9. Coefficients by position relative to a monetary question

Cognitive Endurance as Human Capital (Brown, Kaur, Kingdon, Schofield)

- A field experiment examining whether schooling builds cognitive endurance
- Key insight: Schooling may expand human capital beyond just teaching content
    - It may build our underlying capacity for cognition itself
- Setting: 1,600 primary school students in India
- Method: RCT increasing time in sustained cognitive activity
- Finding: Cognitive practice improves both academic performance and cognitive endurance

Research Question

- **Core Question:** Does the act of sustained effortful thinking build cognitive capacity?
- **Specific Focus:** Cognitive endurance
  - Definition: Ability to sustain performance during cognitively effortful tasks
  - Empirical implication: Performance declines over time during tests/tasks
- **Why it matters:**
  - Productive activity requires sustained mental effort
  - May be a channel through which schooling affects life outcomes
  - Could contribute to inequality if differentially developed

Motivation: Systematic Differences in Cognitive Fatigue

- Authors document performance declines in TIMSS and PISA tests
- Key findings:
  - All students show cognitive fatigue (9-25% decline)
  - Disadvantaged students decline much faster:
    - Poor countries: 3x steeper decline than rich countries
    - US: Black/Hispanic students show 72% steeper decline than White students
- Correlational evidence: Less advantaged students spend less time in independent practice at school

Setting and design

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  - Rote pedagogy, crowded and disrupted classrooms, wide within-class spread in achievement

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- **Three arms:**
  1. Control: Status quo math study hall (minimal practice)
  2. Math Practice: Solve math problems on tablets
  3. Games Practice: Non-academic puzzles (mazes, tangrams)

- **Implementation:**
  - 20-minute sessions, 1-3 times per week
  - August to January (10-20 hours total)
  - During elective periods (no crowding out)

Why Two Treatment Arms?

- **Math Practice:**
  - Mimics what good schooling does
  - Academic content + cognitive practice
  - But conflates learning with endurance building

- **Games Practice:**
  - Pure test of mechanism
  - No academic content (no numbers/words)
  - If effective, shows thinking itself builds capacity

- Both use simple tablet apps to enable individualized difficulty

Data and Outcomes

- **School Performance:**
  - Grades in Hindi, English, Math
  - Mid-year and end-of-year

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  - Listening comprehension
  - Raven’s Matrices (fluid intelligence)
  - Mathematics
  - Question order randomized, ample time given. Why is this important?

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- **Psychology Measures:**
  - Sustained Attention Response Task (SART)
  - Classroom attentiveness ratings

Empirical strategy

Approach 1: fixed location effects. Let performance depend on where a question sits in the test:

$$\text{Correct}_{ils} = \beta_0 + \beta_1 \text{CogPractice}_s + \sum_{l=2}^{10} \delta_l \text{Decile}_l + \beta_2 \text{CogPractice}_s \times \mathbf{1}[\text{Deciles 2-5}]$$

$$+ \beta_3 \text{CogPractice}_s \times \mathbf{1}[\text{Deciles 6-10}] + \beta_4 \text{Baseline}_s + \gamma_{il} + \epsilon_{ils}$$

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Approach 2: predicted decline. Compute the control-group decline at each position, then interact it with treatment. The coefficient gives the **fraction of expected decline the treatment removes** – a data-driven measure rather than an imposed split.

Main Result 1: School Performance

- **Overall effect:** 0.09 SD improvement ($p = 0.01$)

- **By subject:**
  - Hindi: 0.099 SD ($p = 0.012$)
  - English: 0.092 SD ($p = 0.024$)
  - Math: 0.085 SD ($p = 0.025$)

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  - English: 0.092 SD ($p = 0.024$)
  - Math: 0.085 SD ($p = 0.025$)
- **Key finding:** Math and Games arms have similar effects
  - Games improve academic performance despite no academic content
  - Suggests mechanism is building cognitive capacity, not learning

Main Result 2: Cognitive Endurance

- **Control group:** 12% decline from start to end of tests
- **Treatment effect:** Reduces decline by 22% ($p = 0.006$)
  - Math arm: 21.9% reduction ($p = 0.018$)
  - Games arm: 22.0% reduction ($p = 0.015$)
- **Pattern of effects:**
  - No effect at beginning of tests (no level effect)
  - Effects emerge only in second half when fatigue sets in
  - Rules out general confidence/motivation explanations
- **Persistence:** Effects remain 3-5 months post-intervention

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- Corroborated by the psychology measures — sustained attention (SART) and classroom attentiveness both improve

Mechanisms: Ruling Out Alternatives

What other alternatives could explain these results?

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- **Not just motivation:**
  - Cross-randomized incentives for test performance
  - Incentives increase level but don’t reduce decline

- **Not perseverance/grit:**
  - Treatment doesn’t reduce discouragement effects
  - After hard question, still do worse on next question

- **Not other cognitive skills:**
  - No effects at test beginning rules out working memory
  - Pattern specific to fatigue, not general ability

Complementary Evidence: Natural Schooling

- Use birthday cutoffs for school enrollment in Pakistan
- Fuzzy RD comparing students of same age with different current years of schooling
  - What's the key identification assumption here?

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  - It fails if families time births around the cutoff, or manoeuvre children into the year they prefer
- **Results:**
  - Additional year of school reduces decline by 31%
  - Effect 3.4x larger than experimental intervention
  - Concentrated in higher-quality schools
  - Schools with more independent practice show larger effects
- Suggests natural schooling also builds cognitive endurance

Broader Relevance: Adults in the Real World

- **Data entry workers:**
  - 12% increase in error rates from 10am to 4pm
  - Less educated workers decline twice as fast
- **Voters:**
  - More likely to choose default option later on ballot
  - Disadvantaged neighborhoods show 29% steeper decline
- Cognitive endurance matters throughout life
- May perpetuate inequality if differentially developed

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- **Future directions?** Does it persist into the labour market? Does it substitute for or complement content teaching? And what is the cheapest technology that builds it?

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- Then: **test scores** are a poor proxy for learning, in ways that fall hardest on the children we most want to measure
- **Every step of that chain is a substitution we made because we had to — and every one is where the research is**

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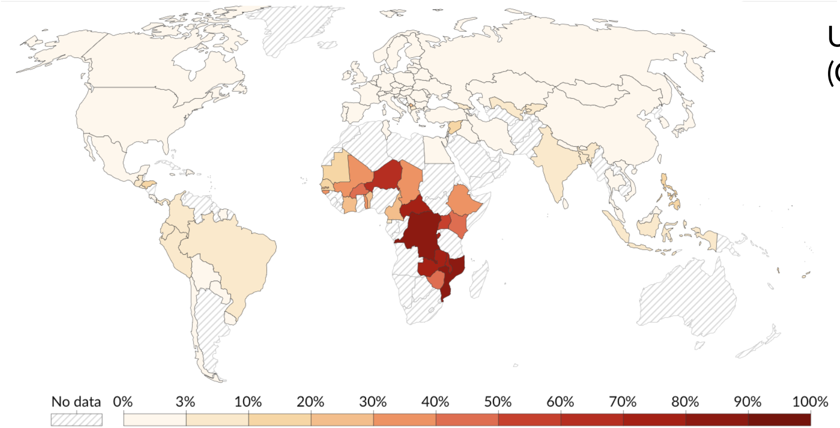
- **Next week**: intrahousehold allocation — what happens when we stop pretending the household is one person

Share of population living in extreme poverty (<$3/day)



United States: 1.25%
(Canada: 0.25%)

Image: OurWorldInData. Data: World Bank.

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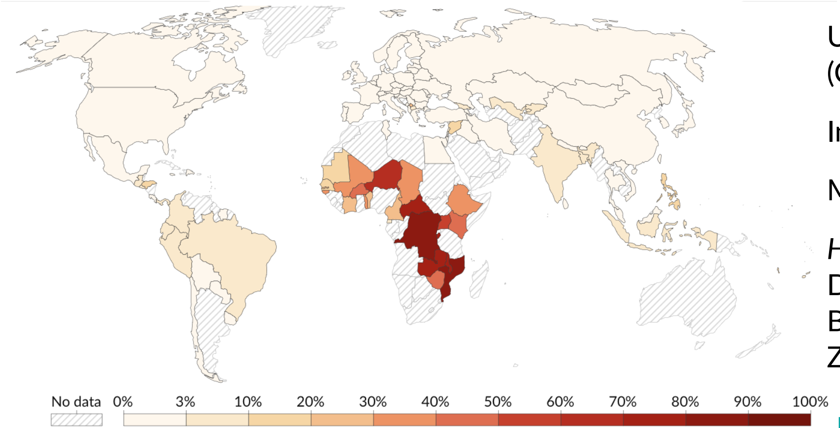

United States: 1.25%
(Canada: 0.25%)

India: 5.25%

Mali: 36%

Highest:
DRC (85%),
Burundi (74%),
Zambia (72%)

Back

Image: OurWorldInData. Data: World Bank.

Average learning-adjusted years of schooling

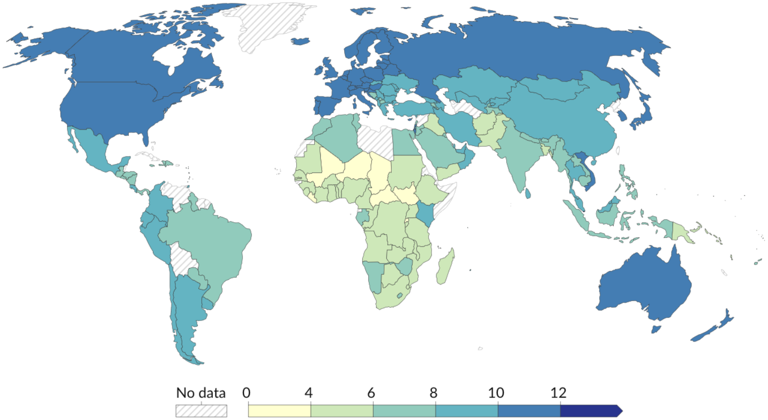

Image: OurWorldInData. Data: World Bank, Filmer et al. 2018.

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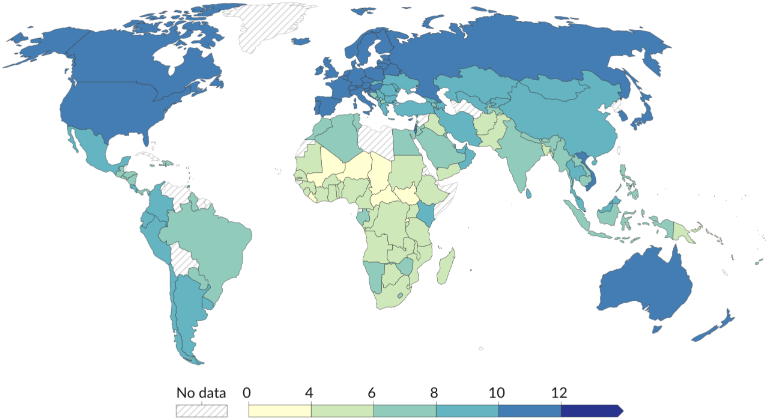

United States:
10.6 years

Image: OurWorldInData. Data: World Bank, Filmer et al. 2018.

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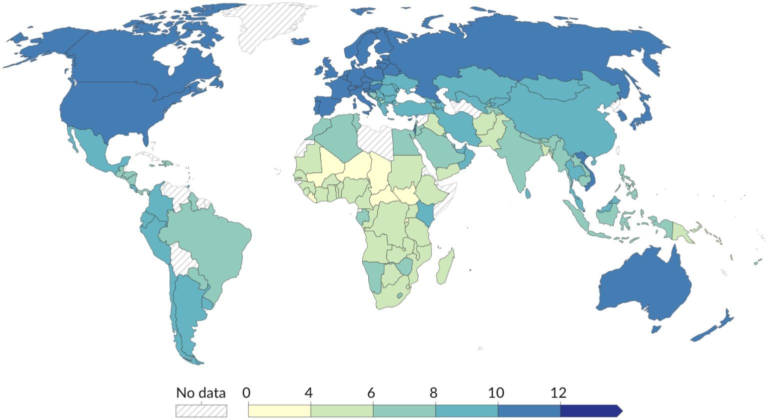


United States: 10.6 years

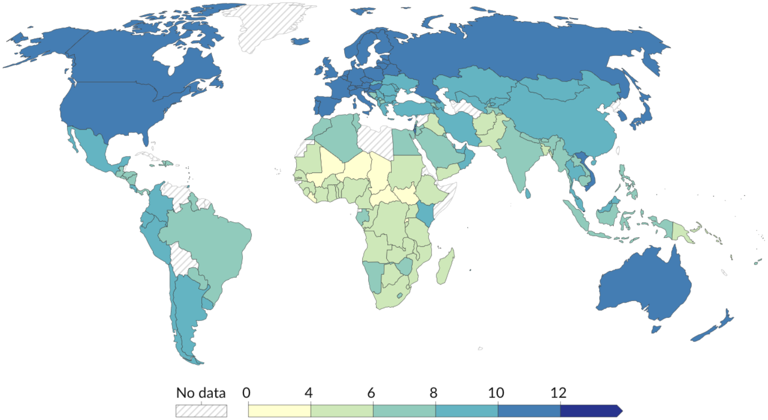
India: 7.1 years

Nigeria: 5 years

Mali: 2.6 years

Image: OurWorldInData. Data: World Bank, Filmer et al. 2018.

Average learning-adjusted years of schooling



United States: 10.6 years

India: 7.1 years

Nigeria: 5 years

Mali: 2.6 years

Vietnam: 10.7 years

Back

Image: OurWorldInData. Data: World Bank, Filmer et al. 2018.

Working hours decrease with income

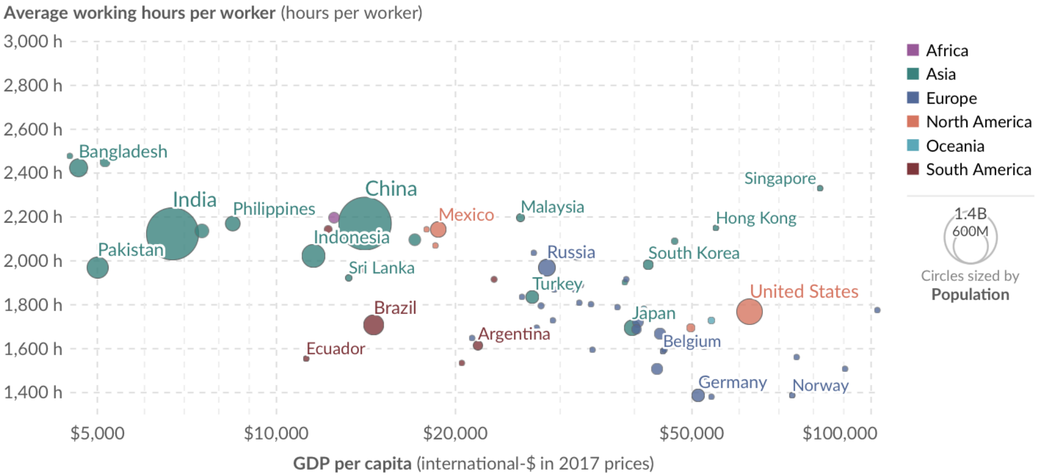
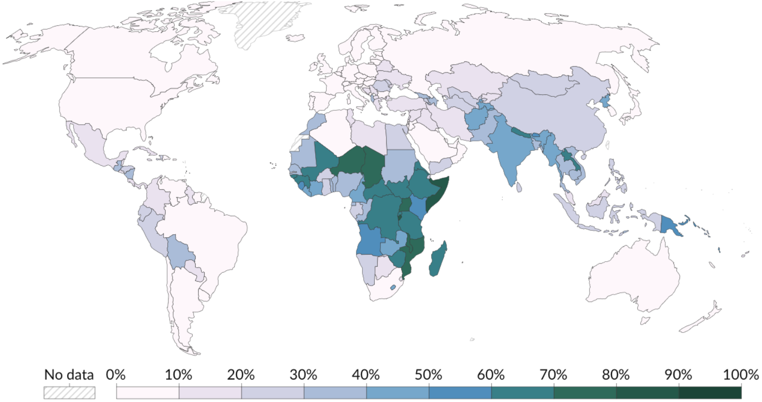


Image: OurWorldInData. Data: Feenstra et al. - Penn World Table (2023).

As countries become richer, workers move out of agriculture



United States: 1%

India: 43%

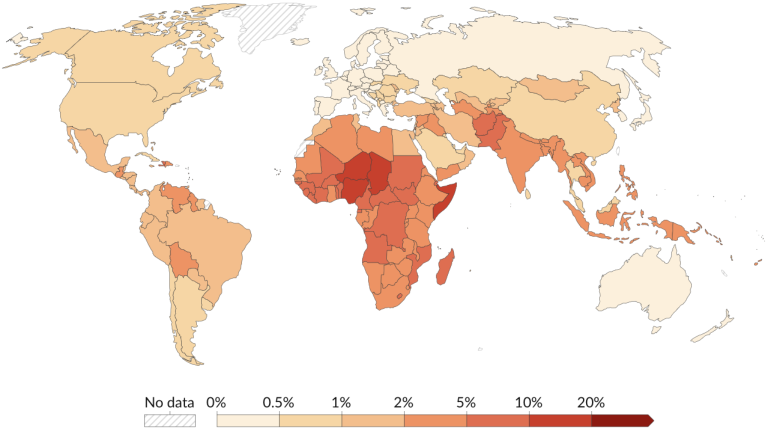
Nigeria: 35%

Somalia: 80%

Back

Image: OurWorldInData. Data: World Bank.

Share of children who die before the age of five



United States: 0.6%

India: 2.8%

Nigeria: 10.5%

Somalia: 10.4%

Back

Image: OurWorldInData. Data: United Nations.

Maternal mortality rate (out of 100,000 live births)

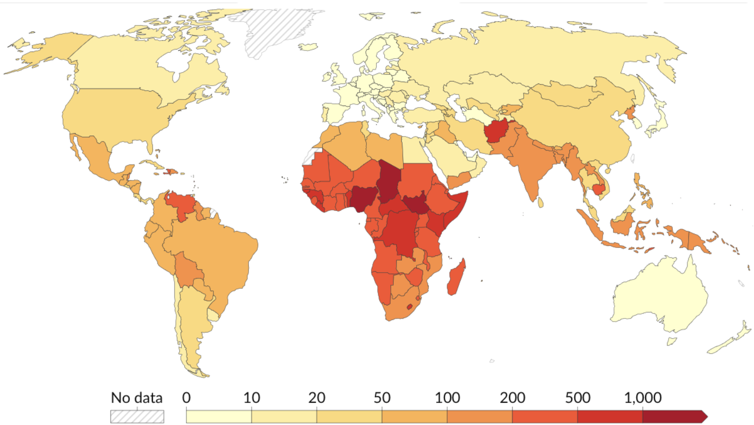


Image: OurWorldInData. Data: United Nations.

US: 21 (Canada: 11, Sweden: 5)

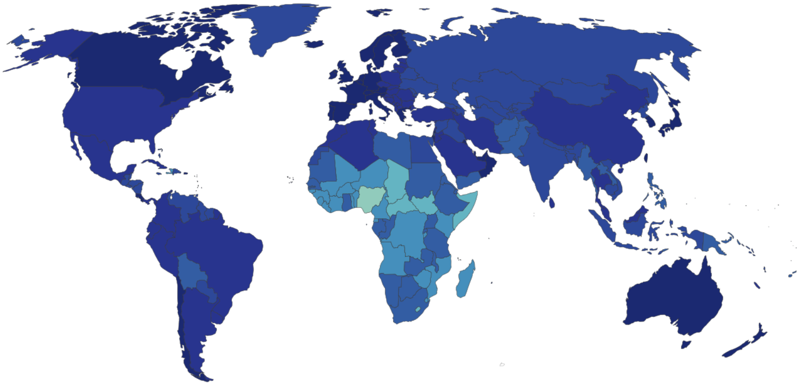
India: 103

Nigeria: 1,047

Somalia: 621

Back

Life Expectancy



US: 79 (Canada: 83, Sweden: 83)

India: 72

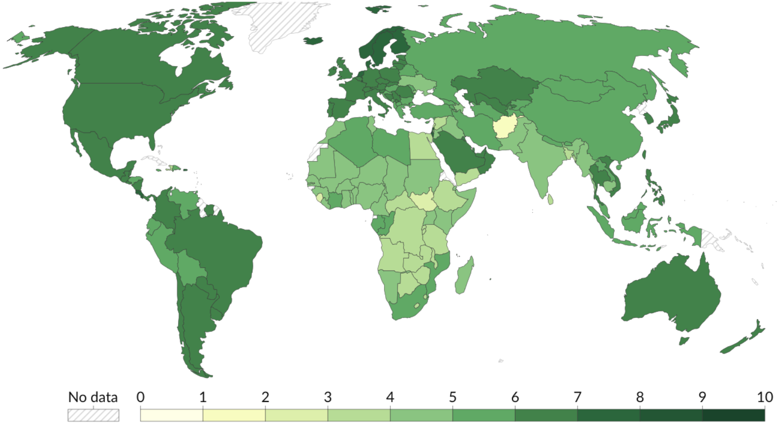
Nigeria: 55

Somalia: 59

Back

Image: OurWorldInData. Data: URiley (2005); Zijdeman et al. (2015); HMD (2024); UN WPP (2024).

Life satisfaction increases with income



On 0-10 scale...

United States: 6.2

India: 4.39

Nigeria: 4.88

Somalia: 4.34

Image: OurWorldInData. Data: Wellbeing Research Centre.

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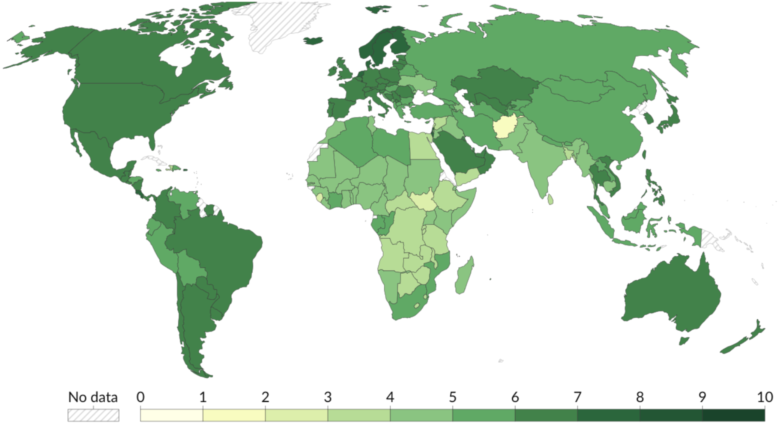

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Iceland (7.51)

Life satisfaction increases with income

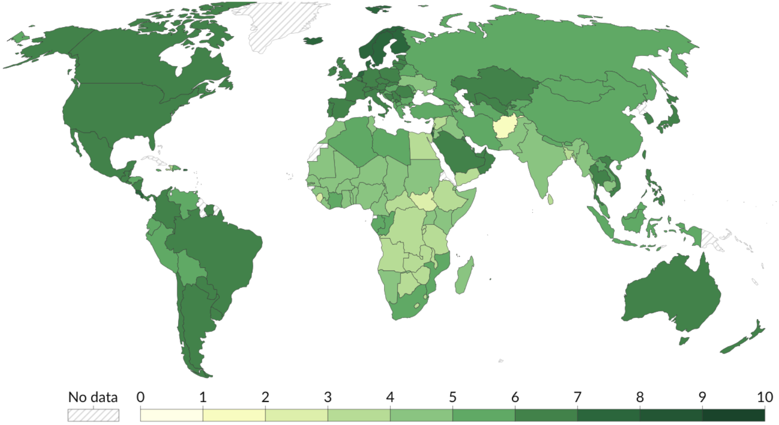

Image: OurWorldInData. Data: Wellbeing Research Centre.

On 0-10 scale...

United States: 6.2

India: 4.39

Nigeria: 4.88

Somalia: 4.34

Highest:
Finland (7.74),
Denmark (7.52),
Iceland (7.51)

Lowest:
Afghanistan (1.36),
South Sudan (2.82),
Sierra Leone (3.0)

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